

Miles Liu

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Generalist software and robotics engineer who works across the whole stack, from sensors and actuators through real-time control to perception, algorithms, integration, and AI tooling around it. Most recently a Systems Engineer at Intuitive Surgical's Advanced Product Development team, where I led software for a micron-precision optical-mechanical sensor alignment system: systems design, control loops, sensor and actuator integration, systems debugging, and the test and diagnostic tooling to drive iterative prototyping. Co-founded Ancora Data and built its AI data automation platform end to end, which gives me current experience shipping production software and AI systems. Fluent in Python and C++, and proficient with AI coding agents.

Education

Johns Hopkins University, Whiting School of Eng. - M.S.E. Robotics & B.S. Mech Eng. **Aug 2017 – May 2022**

- Dean's List, Pi Tau Sigma Honors Society, General Honors; GPA 3.67; Minors: Computer Science, History

Skills

- **Robotics Software:** ROS, robot kinematics, real-time control, sensor fusion, computer vision, behavior trees and state machines, fault detection, safety guardrails, hardware-in-the-loop testing, motion and path planning
- **Hardware and Embedded:** sensors, imaging arrays, actuators, device drivers, FPGAs, fieldbus protocols (EtherCAT, CAN, Ethernet/IP), serial interfaces (I2C, SPI, UART), mechanical design
- **Systems Integration and Test:** system debugging, architecture design, formal verification and validation, FMEA, failure triage, log analysis, cycle time and latency optimization, regression testing
- **Backend and Cloud:** Linux, Python and modern C++ in production, REST APIs (FastAPI), PostgreSQL, MySQL, SQLite, Docker, CI/CD, concurrency and async systems, OAuth and JWT, unit and integration testing
- **AI and ML:** PyTorch (MLP training), ONNX Runtime inference, ML-based diagnostics, exploratory data analysis, LLMs; agent harness, tooling, and testing; RAG, human-in-the-loop workflows, reinforcement learning (PPO, rsl-rl), IsaacLab
- **Languages:** Python, C++, C, SQL, JavaScript and TypeScript, CUDA, MATLAB, NI LabVIEW

Experience

Systems Engineer, Advanced Product Development, Intuitive Surgical **Aug 2022 – Jul 2026**

- Led Software for an advanced micron-precision optical-mechanical alignment system for OFDR sensing technology; built the control stack with **C++/NI LabVIEW/CUDA** on **Linux** interfacing with custom FPGAs and drivers
 - Drove **system-level integration and testing** across cross-disciplinary teams, supporting actuator, electronics, firmware, and mechanical design while owning software end to end; delivered 10 major milestones over 3 years
 - Led software optimization and **reduced alignment routine cycle time from 5+ minutes to 15 seconds**.
 - Led cross-disciplinary debugging and root-cause analysis spanning mechanical, electrical, firmware, and software, validating real-time control and imaging performance. **99.9%** success rate on 1000 repeatability test trials.
- Designed, built, validated, and productionized an ML-powered failure-diagnosis tool for the shape sensor on the Ion platform, built around a neural net trained in PyTorch and ONNX for inference, using logs from hundreds of deployed systems; fully adopted by the Ion service team and used for site analysis, fleet reporting, and business-unit decisions

Co-founder & CTO, Ancora Data

Jul 2025 – Present

- Co-founded and led technical development of an AI-powered data automation platform for back-office workflows.
- Designed and tested an experimental hybrid agentic orchestration and execution engine for flexible data automation tasks, alternating LLM reasoning phases (planning, checkpointing, recovery) with deterministic toolchain execution
- Engaged with 20+ prospective and active clients to scope, design, and forward-deploy custom solutions

Graduate Research Assistant, JHU Computer Integrated Interventional Systems Laboratory

Aug 2021 – Aug 2022

- Parallelized a mosquito salivary gland extraction robot for malaria vaccine production and migrated its codebase from single-threaded scripts to **ROS**, with computer vision, SQL logging, and hardware communication

Lead Product Development Engineer, ClearMask LLC

May 2020 – Jul 2022

- Led test engineering on three FDA Class II medical face-mask products under the CTO, ~5M units sold in 2021

Projects

Spidertron - RL Locomotion for a Custom 18-DOF Hexapod, *In progress, sim only*

[mliu59/hexapod-rl-sim](https://mliu59.github.io/hexapod-rl-sim)

- As a learning project, using RL (goal-conditioned PPO with rsl-rl) with IsaacLab to train a custom 6x3-DoF hexapod robot for locomotion tasks on flat and rugged terrain with obstacles (planned). Currently in progress

Wikispeedruns - Competitive Wikipedia Surfing Browser Game

wikispeedruns.com

- A browser-based online game on Vue + Python/Flask + MySQL, **taken from 0 to 1** with a team of three
- Organically grown to **201k+ registered users, 2M+ games played** over 5+ years of continuous operational ownership